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DBMS LAB 9-TRIGGRES

1)Automatic Maintenance of Student Count:

- `CREATE DATABASE CollegeDBB;`
- `USE CollegeDBB;`
-  `CREATE TABLE Student (
 student_id INT PRIMARY KEY,
 name VARCHAR(100),
 department VARCHAR(50)
);`
-  `CREATE TABLE Student_Count (
 Total_Students INT
);`
- `INSERT INTO Student_Count VALUES (0);`

DELIMITER //

- `CREATE TRIGGER after_student_insert`
`AFTER INSERT ON Student`
`FOR EACH ROW`
`BEGIN`
 `UPDATE Student_Count`
 `SET Total_Students = Total_Students + 1;`
`END //`
- DELIMITER ;
- `INSERT INTO Student (student_id, name, department)`
`VALUES (098, 'Sidharth', 'CS');`
 - `INSERT INTO Student (student_id, name, department)`
`VALUES (065, 'Kiara', 'ECE');`
- `SELECT * FROM Student;`
- `SELECT * FROM Student_Count;`

Result Grid			
Filter Rows:			
	student_id	name	department
▶	65	Kiara	ECE
	98	Sidharth	CS
*	NULL	NULL	NULL

Result Grid	
Filter Rows:	
	Total_Students
▶	2

2)Maintain an Automatic Salary Audit:

```
1 • CREATE DATABASE CompanyDB;
2 • USE CompanyDB;
3
4 • CREATE TABLE Employee (
5     Emp_ID INT PRIMARY KEY,
6     Emp_Name VARCHAR(50),
7     Department VARCHAR(50),
8     Salary DECIMAL(10,2)
9 );
10
11 • CREATE TABLE Salary_Audit (
12     Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
13     Emp_ID INT,
14     Old_Salary DECIMAL(10,2),
15     New_Salary DECIMAL(10,2),
16     Change_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
17 );
18
19 • INSERT INTO Employee (Emp_ID, Emp_Name, Department, Salary) VALUES
20     (108, 'Sanjana', 'Engineering', 57000.00),
21     (107, 'Shrujana', 'Marketing', 45000.00);
22
23 • SELECT * FROM Employee;
```

DELIMITER //

- ```
CREATE TRIGGER after_salary_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
 IF OLD.Salary != NEW.Salary THEN
 INSERT INTO Salary_Audit (Emp_ID, Old_Salary, New_Salary)
 VALUES (OLD.Emp_ID, OLD.Salary, NEW.Salary);
 END IF;
END //
```

DELIMITER ;

- ```
UPDATE Employee
SET Salary = 65000.00
WHERE Emp_ID = 108;
```
- ```
UPDATE Employee
SET Salary = 52000.00
WHERE Emp_ID = 107;
```
- ```
SELECT * FROM Employee;
```

	Emp_ID	Emp_Name	Department	Salary
▶	107	Shrujana	Marketing	45000.00
	108	Sanjana	Engineering	57000.00
*	NULL	NULL	NULL	NULL

Result Grid				Filter Rows:	Edit:
	Emp_ID	Emp_Name	Department	Salary	
▶	107	Shrujana	Marketing	52000.00	
	108	Sanjana	Engineering	65000.00	
*	NULL	NULL	NULL	NULL	

Result Grid				Filter Rows:		Edit:				Exp
	Audit_ID	Emp_ID	Old_Salary	New_Salary	Change_Time					
▶	1	108	57000.00	65000.00	2026-09-29 12:00:27					
	2	107	45000.00	52000.00	2026-09-29 12:00:27					
*	NULL	NULL	NULL	NULL	NULL					

3)Prevent Deletion of Important Employees:

```
1 • CREATE DATABASE CompanyDB2;
2 • USE CompanyDB2;
3
4 • CREATE TABLE Employee (
5     Emp_ID INT PRIMARY KEY,
6     Emp_Name VARCHAR(50),
7     Department VARCHAR(50),
8     Salary DECIMAL(10,2)
9 );
10
11 • CREATE TABLE Deleted_Employee_Log (
12     Log_ID INT AUTO_INCREMENT PRIMARY KEY,
13     Emp_ID INT,
14     Emp_Name VARCHAR(50),
15     Deleted_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
16 );
17
18 • INSERT INTO Employee VALUES (101, 'Conrad', 'IT', 75000);
19 • INSERT INTO Employee VALUES (102, 'Garret', 'HR', 50000);
20 • INSERT INTO Employee VALUES (103, 'Sidharth', 'Management', 85000);
21 • INSERT INTO Employee VALUES (104, 'Kiara', 'Finance', 65000);
22
```

```

DELIMITER //

CREATE TRIGGER before_employee_delete
BEFORE DELETE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.Department = 'Management' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Management employees cannot be deleted';
    END IF;
END //

CREATE TRIGGER after_employee_delete
AFTER DELETE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Deleted_Employee_Log (Emp_ID, Emp_Name)
    VALUES (OLD.Emp_ID, OLD.Emp_Name);
END //

DELIMITER ;

DELETE FROM Employee WHERE Emp_ID = 101;

DELETE FROM Employee WHERE Emp_ID = 103;

SELECT * FROM Employee;

SELECT * FROM Deleted_Employee_Log;

```

->Shows error as we cannot delete management employees

✓	43 12:07:27	CREATE TRIGGER before_employee_delete BEFORE DELETE ON Employee FOR EACH ROW BEGIN	1 row(s) affected
✓	44 12:07:27	DELETE FROM Employee WHERE Emp_ID = 101	1 row(s) affected
✗	45 12:07:27	DELETE FROM Employee WHERE Emp_ID = 103	Error Code: 1644. Management employees cannot be deleted
